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Algebra

14

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ANSWER KEY · SAT QUESTION BANK

2026-05-29

Q1

B

B. $7x + 18y = 630$

Choice B is correct. A line in the xy-plane can be written as $y = mx + b$, where m is the slope of the line and b is the y-coordinate of the y-intercept. The graph shown is a line passing through the points $(0, 35)$ and $(90, 0)$. Substituting 0 for x and 35 for y in the equation $y = mx + b$ yields $35 = m(0) + b$, or $35 = b$. Substituting 35 for b, 90 for x, and 0 for y in the equation $y = mx + b$ yields $0 = 90m + 35$. Subtracting 35 from both sides of this equation yields $-35 = 90m$. Dividing both sides of this equation by 90 yields $-\frac{35}{90} = m$, or $-\frac{7}{18} = m$. Substituting $-\frac{7}{18}$ for m and 35 for b in the equation $y = mx + b$ yields $y = -\frac{7}{18}x + 35$. Multiplying both sides of this equation by 18 yields $18y = -7x + 35(18)$, or $18y = -7x + 630$. Adding $7x$ to both sides of this equation yields $7x + 18y = 630$. Therefore, the equation $7x + 18y = 630$ represents the relationship between x and y modeled by the graph.

Choice A is incorrect. The point $(0, 35)$ is not on the graph of this equation, since $7(0) + 18 = 18$, not 35.

Choice C is incorrect. The point $(0, 35)$ is not on the graph of this equation, since $18(0) + 7 = 7$, not 35.

Choice D is incorrect. The point $(90, 0)$ is not on the graph of this equation, since $18(90) + 7(0) = 1{,}620$, not 630.

Q2 A

A. The monthly sales revenue, in dollars, from selling x tape dispensers

Choice A is correct. It's given that the equation $15{,}000 = 2.00x - 4{,}500$ represents this situation for a month where x tape dispensers are created and sold. It's also given that the company calculates its monthly profit, in dollars, by subtracting its fixed monthly costs, in dollars, from its monthly sales revenue, in dollars. It follows that $2.00x$ represents the monthly sales revenue, in dollars. Therefore, the best interpretation of $2.00x$ in this context is the monthly sales revenue from selling x tape dispensers.

Choice B is incorrect. This is the best interpretation of 2.00, not $2.00x$.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect. This is the best interpretation of $4{,}500$, not $2.00x$.

Q3 A

A. The team will have moved all the cargo in about 4.8 hours.

Choice A is correct. The x-intercept of the line with equation $y = 120 - 25x$ can be found by substituting 0 for y and finding the value of x. When $y = 0$, $x = 4.8$, so the x-intercept is at $(4.8, 0)$. Since y represents the number of tons of cargo remaining to be moved x hours after the team started working, it follows that the x-intercept refers to the team having no cargo remaining to be moved after 4.8 hours. In other words, the team will have moved all of the cargo after about 4.8 hours.

Choice B is incorrect and may result from incorrectly interpreting the value 4.8. Choices C and D are incorrect and may result from misunderstanding the x-intercept. These statements are accurate but not directly relevant to the x-intercept.

Q4

A

A. $y = 6x + 3$
$y = 6x + 9$

Choice A is correct. A system of two linear equations in two variables, x and y, has no solution if the graphs of the lines represented by the equations in the xy-plane are distinct and parallel. The graphs of two lines in the xy-plane represented by equations in slope-intercept form, $y = mx + b$, where m and b are constants, are parallel if their slopes, m, are the same and are distinct if their y-coordinates of the y-intercepts, b, are different. In the equations $y = 6x + 3$ and $y = 6x + 9$, the values of m are each 6, and the values of b are 3 and 9, respectively. Since the slopes of these lines are the same and the y-coordinates of the y-intercepts are different, it follows that the system of linear equations in choice A has no solution.

Choice B is incorrect. The two lines represented by these equations are a horizontal line and a line with a slope of 10 that have the same y-coordinate of the y-intercept. Therefore, this system has a solution, $0, 10$, rather than no solution.

Choice C is incorrect. The two lines represented by these equations have different slopes and the same y-coordinate of the y-intercept. Therefore, this system has a solution, $0, 14$, rather than no solution.

Choice D is incorrect. The two lines represented by these equations are a vertical line and a horizontal line. Therefore, this system has a solution, $3, 10$, rather than no solution.

Q5

D

D. $(0, -1)$

Choice D is correct. The solutions to the given system of inequalities is the set of all ordered pairs (x, y) that satisfy both inequalities in the system. For an ordered pair to satisfy the inequality $y \leq x$, the value of the ordered pair's y-coordinate must be less than or equal to the value of the ordered pair's x-coordinate. This is true of the ordered pair $(0, -1)$, because $-1 \leq 0$. To satisfy the inequality $y \leq -x$, the value of the ordered pair's y-coordinate must be less than or equal to the value of the additive inverse of the ordered pair's x-coordinate. This is also true of the ordered pair $(0, -1)$. Because 0 is its own additive inverse, $-1 \leq -(0)$ is the same as $-1 \leq 0$. Therefore, the ordered pair $(0, -1)$ is a solution to the given system of inequalities.

Choice A is incorrect. This ordered pair satisfies only the inequality $y \leq x$ in the given system, not both inequalities. Choice B incorrect. This ordered pair satisfies only the inequality $y \leq -x$ in the system, but not both inequalities. Choice C is incorrect. This ordered pair satisfies neither inequality.

Q6

A

A. $-\frac{92}{3}$

Choice A is correct. The slope, m, of a line in the xy-plane can be found using two points on the line, x_1, y_1 and x_2, y_2, and the slope formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. Based on the given table, the line representing the relationship between x and y in the xy-plane passes through the points $-6, n + 184$, $-3, n + 92$, and $0, n$, where n is a constant. Substituting two of these points, $-3, n + 92$ and $0, n$, for x_1, y_1 and x_2, y_2, respectively, in the slope formula yields $m = \frac{n - n + 92}{0 - -3}$, which is equivalent to $m = \frac{n - n - 92}{0 + 3}$, or $m = -\frac{92}{3}$. Therefore, the slope of the line that represents this relationship in the xy-plane is $-\frac{92}{3}$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q7

241

The correct answer is 241. For a certain animal, it's given that a model predicts the animal weighed 241 pounds when it was born and gained 3 pounds per day in its first year of life. It's also given that this model is defined by an equation in the form $f x = a + bx$, where $f x$ is the predicted weight, in pounds, of the animal x days after it was born, and a and b are constants. It follows that a represents the predicted weight, in pounds, of the animal when it was born and b represents the predicted rate of weight gain, in pounds per day, in its first year of life. Thus, the value of a is 241.

Q8

C

C. 19

Choice C is correct. Let x represent the number of 2-person tents and let y represent the number of 4-person tents. It is given that the total number of tents was 60 and the total number of people in the group was 202. This situation can be expressed as a system of two equations, $x+y=60$ and $2x+4y=202$. The first equation can be rewritten as $y=-x+60$. Substituting $-x+60$ for y in the equation $2x+4y=202$ yields $2x+4(-x+60)=202$. Distributing and combining like terms gives $-2x+240=202$. Subtracting 240 from both sides of $-2x+240=202$ and then dividing both sides by -2 gives $x=19$. Therefore, the number of 2-person tents is 19.

Alternate approach: If each of the 60 tents held 4 people, the total number of people that could be accommodated in tents would be 240. However, the actual number of people who slept in tents was 202. The difference of 38 accounts for the 2-person tents. Since each of these tents holds 2 people fewer than a 4-person tent, $\frac{38}{2}=19$ gives the number of 2-person tents.

Choice A is incorrect. This choice may result from assuming exactly half of the tents hold 2 people. If that were true, then the total number of people who slept in tents would be $2(30)+4(30)=180$; however, the total number of people who slept in tents was 202, not 180. Choice B is incorrect. If 20 tents were 2-person tents, then the remaining 40 tents would be 4-person tents. Since all the tents were filled to capacity, the total number of people who slept in tents would be $2(20)+4(40)=40+160=200$; however, the total number of people who slept in tents was 202, not 200. Choice D is incorrect. If 18 tents were 2-person tents, then the remaining 42 tents would be 4-person tents. Since all the tents were filled to capacity, the total number of people who slept in tents would be $2(18)+4(42)=36+168=204$; however, the total number of people who slept in tents was 202, not 204.

Q9

14.66

Accept also: `14.67`, `44/3`

The correct answer is $\frac{44}{3}$. A linear equation can be written in the form $y = mx + b$, where m is the slope of the graph of the equation in the xy-plane and $0, b$ is the y-intercept. Distributing the $\frac{1}{3}$ in the equation $y = \frac{1}{3}29x + 10 + 5x$ yields $y = \frac{29}{3}x + \frac{10}{3} + 5x$. Combining like terms on the right-hand side of this equation yields $y = \frac{44}{3}x + \frac{10}{3}$. This equation is in the form $y = mx + b$, where $m = \frac{44}{3}$ and $b = \frac{10}{3}$. Therefore, the slope of the graph of the given equation in the xy-plane is $\frac{44}{3}$. Note that 44/3, 14.66, and 14.67 are examples of ways to enter a correct answer.

Q10

7

The correct answer is 7. An equation of a line in the xy-plane can be written in the form $y = mx + b$, where m is the slope of the line and $(0, b)$ is the y-intercept of the line. It's given that line k has a slope of 5 and a y-intercept of $(0, -35)$. Therefore, $m = 5$ and $b = -35$. Substituting 5 for m and -35 for b in the equation $y = mx + b$ yields $y = 5x - 35$. The x-intercept of a line in the xy-plane is the point where the line intersects the x-axis, which is a point with a y-coordinate of 0. Substituting 0 for y in the equation $y = 5x - 35$ yields $0 = 5x - 35$. Adding 35 to both sides of this equation yields $35 = 5x$. Dividing both sides of this equation by 5 yields $7 = x$. Therefore, the x-coordinate of the x-intercept of line k is 7.

Notes

Accept equivalent forms (e.g. `0.25` ≡ `1/4`). For grid-in items, decimal and fraction answers are both valid.

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